

Functional Extensionality and Its Discontents. The Curry–Howard–Lambek Correspondence and Natural Language Semantics

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Abstract. What does it mean for two expressions to have the same meaning? In formal semantics, the standard answer is functionally extensional: two expressions mean the same if their functional outputs are equal on all possible inputs. This answer, we argue, is not merely a methodological choice but a consequence of the formal foundations most semantic theories rely on. The Curry-Howard-Lambek correspondence (CHLC), unifying logic, computation, and category theory, carries an implicit commitment to functional extensionality that propagates into the semantic analysis of natural language. The problem is that language does not play by these rules. Speakers routinely distinguish meanings that extensional criteria cannot capture immediately. Distinctions may resist reduction to truth conditions and thus necessitate additional or more complex input parameters. We argue that this is not a peripheral puzzle but a structural mismatch between the formal language and the object language it describes. The remedy comes from using the category theoretic dimension of CHLC that has so far been neglected. Grounding meaning in the relational Yoneda perspective shifts the notion of identity from one of equal outputs to one of structural position: an expression means what it does by virtue of how it relates to everything else.

Keywords: Curry-Howard-Lambek Correspondence · Extensionality · Intensionality · Hyperintensionality · Yoneda Lemma.

1 Curry-Howard-Lambek in Natural Language Semantics

The Curry–Howard–Lambek correspondence (CHLC) provides a foundational connection between logic, computation, and category theory. Propositions correspond to types, programs to proofs, and both find a category theoretic interpretation in cartesian closed categories (CCCs), cf. [14,15]. In logic and computer science, CHLC and its implications have been extensively studied. In contrast, its significance for natural language semantics remains remarkably unacknowledged, especially with respect to the distinction of extensionality and intensionality. Although there are no explicit discussions of the relation between natural language

semantics and CHLC, there are subtle, rather implicit reflections of it in works on and applications of type theory, comprising two major strands.¹ Rich(er) typed theories (RTTs), cf. [7,6], following Martin-Löf’s intuitionistic type theory [19], exemplify the direction ‘from propositions to types’, while simple type theories (STTs), originating in Church’s typed λ -calculus [4] and developed in Montague grammar [23], assume a set of basic types to construct propositions. Hence, natural language semantics has implicitly been focused exclusively on the Curry-Howard side of CHLC as the guiding perspective on compositional meaning. Thus, formal semantics exhibits a notable gap regarding the category theoretic Lambek side of CHLC.²

First, we argue that this asymmetry is not merely historical or accidental, but reflects an implicit, yet crucial commitment in semantics to functional extensionality inherited from CHLC itself. Although this assumption has not been questioned previously and seems natural and convincing, potentially even trivial, we argue that this seemingly harmless assumption results in potentially objectionable design choices for a parsimonious meaning analysis and should thus be dropped. We therefore identify implications of extensionality in CHLC (section 2.1), especially for natural language (section 2.2). Then we discuss the consequences of a non-extensional version of CHLC (section 2.3). In particular, we claim that a non fully extensional variant of CHLC is a better match for semantics, as language does not fully adhere to extensionality. Thus, giving up on strict extensionality creates a better alignment between formal and object language. Second, we argue that possible drawbacks of a non fully extensional CHLC can naturally be answered by a semantics fully couched in the category theoretic side of CHLC (section 3.1). Specifically, analyzing meaning category theoretically motivates a shift from element-based to structural notions of identity, induced by the inherent relational perspective in category theory. The Yoneda perspective makes this structural position formally explicit by showing how objects are defined through the context of morphisms in which they participate rather than relying on exact identity of objects. Thus, truth and meaning are reconceived in structural, epistemic terms rather than as mere outputs of extensional functions (section 3.2), as the category theoretic perspective creates an intensional view of (semantic) functions and identity while still providing a uniform notion of compositionality. We conclude with the claim that even constraints on meaning can be consistently described in category theoretic terms, thereby allowing to treat meaning and restrictions on it with the same formal language (section 3.3).

2 Extensional Bias

2.1 CHLC and Extensionality

In standard extensional settings such calculus, many programming languages, or extensional meaning predicates, Leibniz principle of equality strictly applies. The

¹ See [29] for a more detailed overview.

² Cf. [27] for first steps towards a comprehensive approach to natural language meaning from a category theoretic perspective.

principle, often termed the 'indiscernibility of identicals', states that identical entities have the exact same properties. As entities act as inputs of functions and thereby determine their output, the principle also ensures well-definedness of functions, cf. [10,13]: 'Equal inputs of functions yield equal outputs'.

- (1) If $x = y$ then $\forall f : f(x) = f(y)$.

Functional extensionality can be understood as the application of Leibniz' principle on functions themselves: Functions are indiscernible if they yield equal outputs on any equal input. In his work [9] from 1930, Curry worked out the consequences of restricting a Schönfinkel algebra, cf. [28], with the additional requirement of functional extensionality, given in Lambek's notation, cf. [15]:

- (2) If $f'x = g'x$ in $\mathcal{A}[x]$ then $f = g$ in $\mathcal{A}$.

This restricts general Schönfinkel algebras to Curry algebras, which are known to be functionally complete due to the functional extensionality (2). Lambek formulates this result as follows:

Proposition 1 (Functional completeness of Curry algebras). *Over a Curry algebra $\mathcal{A}$ every polynomial $\varphi(x)$ may be uniquely written in the form $f'x$ with $f \in |\mathcal{A}|$.*

As CHLC provides the fundamental proof of the isomorphism between combinatory logic, λ -calculus, and category theory, the functional completeness in Curry algebras naturally translates into the category theoretic equivalent, CCCs:

Proposition 2 (Functional completeness of CCCs). *For every polynomial $\varphi(x) : B \rightarrow C$ in an indeterminate $x : 1 \rightarrow A$ there is a unique constant $f : A \times B \rightarrow C$ such that $f \circ \langle x \circ 0_B, 1_B \rangle = \varphi(x)$.*

Consequently, not only does CHLC prove the equivalence of certain types, proofs, and categories, but it also states that these domains are extensional in precisely the same sense, which has not been made explicit so far. Notably, in all domains functional extensionality guarantees the uniqueness of the corresponding elements, i.e. categorical objects and morphisms, (function) types, and proofs. Similarly, in natural language semantics, the extensionality of functions (2) implicitly carries over from CHLC into both STTs and RTTs, and has not been questioned to date. In linguistics, the relevant equality is traditionally understood as meaning terms being co-extensional. Thus, co-extensional expressions are expected to evaluate to the same truth value under any predicate. We use a simple typed λ -calculus as an example to illustrate the effects of the core assumption in (2), as STTs probably represent the most widely used domain of CHLC in formal semantics.³ In standard STTs, all meaning terms can be understood and modeled as functions of the appropriate type. Basic types include entities e and

³ Due to CHLC, the subsequent considerations isomorphically apply to RTTs, by relating the increase of input parameters with an increase in fine-grainedness of propositional types. For a more detailed discussion of RTTs, cf. [27].

truth values t (constants, i.e., zero-valued functions), and the predicates over these encompass complex types such as $\langle e, t \rangle$ or $\langle e, \langle e, t \rangle \rangle$.⁴ The compositionality of meaning terms is formally expressed by function composition, and truth, represented by the type t , is essentially identified with evaluated outputs by means of function application (i.e., for λ -calculus, β -reduction). This setup thus naturally respects (2), exemplifying Leibniz principle on λ -expressions p, q : If the output of functions (i.e., meaning terms) p and q is equal (i.e., co-extensional) on all inputs in the model, then $p = q$, known as $\beta\eta$ -equality.

However, as is well known, the analysis falls short by identifying meaning purely by co-extensional outputs of functions over e and t , because all true (resp. all false) propositions would then predicted to be equal on the grounds of equality being co-extensionality. Propositions built up by very different meaning terms would collapse into a single value of type t , not formally representing the Fregean distinction of sense and reference. For that reason, an additional atomic type s and the corresponding intensional function type with $\langle s, \alpha \rangle$ (for any type α) are introduced. Functionally, s is usually represented by an additional input parameter λw , which, following Lewis [17], now identifies the extension of a proposition p with the set of worlds w in which p is true (and the intension of p with a function from that set of worlds into t). The system is then able to discern previously co-extensional expressions (i.e., propositions with the same truth value) and, notably, is still functionally extensional, and thus obeying (2) coherently. It is crucial to note that the 'intensional' distinction of previously equal expressions was captured by an additional input parameter (here: w) that feeds into the extensional function expressing a meaning term. However, even then there are different meaning terms that are predicted as equal, as they all yield the same set of worlds when inputs are equal (i.e., again, co-extensional).

- (3) a. $\lambda w. \text{Hesperus}() = \lambda w. \text{Phosphorus}()$
- b. $\lambda w \lambda x_e. \text{bachelor}(x) = \lambda w \lambda x_e. \text{unmarried}(x) \wedge \text{male}(x)$
- c. $\lambda w \lambda x_e \lambda y_e. \text{eat}(x, y) = \lambda w \lambda x_e \lambda y_e. \text{devour}(x, y)$
- d. $\lambda w \lambda x_e \lambda y_e \lambda z_e. \text{buy}(x_e, y_e, z_e) = \lambda w \lambda x_e \lambda y_e \lambda z_e. \text{sell}(z, x, y)$

It is widely known that natural language meaning is systematically able to draw distinctions that are more fine-grained and cannot be reduced to equality based solely on co-extensional notions, even when we identify expressions as dependent on possible worlds. (Hyper-)Intensional phenomena suggest that meaning is sensitive not only to what is true, but also to how and why something is true.

2.2 The Extensional-Intensional Clash

As has been noted previously (cf., e.g., [25], [16], [1], [6]), identifying the meaning of propositions as the output of extensional functions with a set of possible worlds

⁴ Generally: If α, β are types, then $\langle \alpha, \beta \rangle$ is also a type, and can be seen as a function from a domain of type α to a range of type β .

is not fine-grained enough for the various distinctions that speakers can naturally and easily discriminate.⁵

- (4) a. Mary Jane loves Peter Parker.
b. Mary Jane loves Spiderman.
- (5) a. Jim knows Phil is a groundhog.
b. Jim knows Phil is a woodchuck.
- (6) a. John believes that every prime number is divisible only by itself and one.
b. John believes that if $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- (7) CONTEXT: Theseus' starship flies through the galaxy and exchanges all of its parts during travel due to repair work. Watto, a junk dealer, quickly detects the repairs and does not recognize the identity of the ship that just arrived on Tatooine. The Jedi Obi-Wan is simply happy about the return of Theseus and his crew. He recognizes the identity of the ship.
a. Watto correctly believes that Theseus' starship did not arrive.
b. Obi-Wan correctly believes that Theseus' starship arrived.
- (8) CONTEXT: The sociologist Avery and the doctor Robin are discussing what it means to be female.⁶
a. Avery thinks that Imane Khelif is female.
b. Robin thinks that Imane Khelif is female.

Although on the surface, (4)-(6) have different meaning terms in the sentences a. and b. as functional components, their extensional wholes, the propositions, i.e. sets of possible worlds, all collapse into the same denotation, their co-extensionality resp. predicts a. and b. to be equal.⁷ In contrast, while in both (7) and (8) the relevant meaning terms *Theseus' starship* and *female* seem to be identical in the a. and b. versions, speakers most likely feel inclined to distinguish their meaning in the given contexts.^{8,9}

If we consider semantics to be a comprehensive and thorough study of meaning,

⁵ Examples (4), (5), and (6) are cited from [1], [25], and [16] resp.

⁶ The featured Imane Khelif is an Algerian boxer, who won a gold medal in the 2024 Olympics in Paris. Khelif is one of many examples of people in sports whose gender, and therefore the category in which they compete, has been questioned publicly.

⁷ Notably, example (6) is also reminiscent of the problem of logical omniscience. As tautologies all have the same extension $\mathcal{W}$, the set of all possible worlds, and as *believe* is monotonic just like *assume*, *fact*, or *story*, believing any single true proposition would entail believing all true propositions, contrary to intuition. Similarly, the analog holds for contradictions, with all contradictions being co-extensional.

⁸ Note that for (8), the same apparent contradiction as in (7) could arise. One could simultaneously think Khelif to be female 'socially', but not 'medically', or the other way around. Additionally, even in the medical sense, there might be (and are indeed) multiple criteria to use for assigning a sex to people.

⁹ While at first glance examples like (7) might be argued to be artificial and irrelevant to everyday language use, it patterns with (8), which represents how we constantly

including potentially very subtle differences in it, we must acknowledge that distinguishing extensional and intensional dimensions is not enough. Examples (4) to (8) show variants of additional, hyperintensional, dimensions of meaning. It is outside of the scope of this paper to go into the details of different variants of hyperintensionality or to discuss previous approaches to it.¹⁰ Instead, we point out the central underlying logic that all approaches assuming (2) are bound to exhibit, which was already mirrored in the procedure of adding the intensional type s and the corresponding λw . The core idea is this: When expressions are judged equal, i.e., co-extensional, but need to be distinguished with a finer granularity, additional variables are introduced as input of the semantic functions to ensure that meaning distinctions are only made against the backdrop of the functional extensionality assumption all approaches falling under CHLC adhere to. This is apparent in both equality being defined as co-extensionality with respect to truth (ex.: all true/false propositions) or to the intensional version of it, a set of worlds (ex. (4) to (8)). As functions are essentially taken to be extensional, this procedure of adding input parameters actually represents the consistent, indeed necessary consequence for every possible additional parameter meaning is sensitive to. Thereby, the complexity of the phenomenon is always shifted towards additional complexity in the input of the semantic functions, such that the (implicit) assumption (2) is not violated. On the one hand, the meaning functions then always represent the ‘worst case’ scenario, as their definition always incorporates all possible input parameters for reasons of well-definedness. On the other hand, it might not even be possible to define the functions in the first place, as there are potentially infinitely many sensitivities to context, cf. [5, 344]. Moreover, the system dilutes differences between proper semantic arguments and contextual indices, treating them as functional inputs alike. As notably all examples bearing the possibility to distinguish potentially very fine-grained meaning differences feature intensional contexts, the identification of meaning of a (compositional) term solely on the grounds of equal, i.e., co-extensional, outputs of extensional functions, clashes with this intensionality itself and the many interpretations it produces and allows. Thus, it is straightforward to conclude that any extensional semantic analysis is forced to add (infinitely) additional or more complex input variables to account for the various differences in meaning that semantic computation contexts can cause.¹¹ In contrast, the variety of fine-grained distinctions speakers draw does not pose undue interpretive difficulties. We are accustomed to situating meaning computations in their contexts, whereas extensional functions by definition cannot include the context in which they are evaluated. In conclusion, for all approaches assum-

discuss the semantics of expressions, implicitly and explicitly. Hyperintensionality thus not only concerns nominals, but potentially all natural language expressions.

¹⁰ Cf. [2] for a general overview.

¹¹ Treatments in RTTs like [5,6,7], exemplifying intuitionistic type theory, do not add additional input variables, but of increasing complexity through (potentially infinitely) fine-grained types capturing the additional parameter. The mechanism thus also follows the logic of shifting the complexity towards the input of functions.

ing (2), the semantic analysis of (hyper-)intensionality becomes unnecessarily cumbersome, precisely because it clashes with the extensionality built into its framework. Object and formal language inherently adhere to different modes of computation: the intensional phenomenon clashes with its extensional treatment.

2.3 CHLC Without Extensionality

Revisiting the notion of functional completeness and its implications for the correspondence between Curry algebra elements and morphisms in CCCs, we observe that only by assuming (2) all morphisms are ensured to be **uniquely** determined by their behavior on arguments. In other words, two elements of the algebra are taken to be identical if they yield identical results for all inputs. In the category theoretic domain, this corresponds to the identification of morphisms with functions in the classical extensional sense. It is crucial to observe that this understanding of morphisms as functions, functional completeness in CCCs, cf. Prop. 2, does not constitute an absolute property, but hinges on two distinct features, namely representability on the one hand and the assumption of functional extensionality (2) on the other. Importantly, (2) does not entail representability, and neither does representability entail (2). Neither feature alone entails functional completeness: the two components are independent. Algebraically, the representability in CHLC is understood as follows.

Proposition 3 (Representability¹²). *Every polynomial $\varphi(x)$ over a Schönfinkel algebra $\mathcal{A}$ can be written in the form $f'x$, where $f \in \mathcal{A}$.*

It is the additional functional extensionality assumption that strengthens representability to functional completeness by ensuring the uniqueness of the algebraic representations. By CHLC, the same applies to morphisms in CCCs or λ -expressions that uniquely represent a polynomial in $\mathcal{A}$ only by assuming (2). This distinction between representability as such and functional completeness (constituting unique representability) is particularly relevant when we move from purely logical systems to natural language. Here, as we have seen, functions representing meaning terms are often not fully determined by their inputs alone. Two co-extensional expressions may yield identical truth conditions while differing in meaning due to conceptual, epistemic, or explanatory structure. Speakers can easily distinguish expressions with different internal structures or methods of computation, which would analytically predicted to be equal due to their functions evaluating to co-extensional outputs. Although this might not be necessary for functions in calculus or programming,¹³ for evaluating functions expressing co-extensional meaning (such as in (3)) in intensional domains, it was

¹² For statement, proof and further discussion cf. Prop. 1 and explanations in [15].

¹³ Cases like in (i) constitute basic examples:

- (i) a. $4 = 2 + 2 = 2^2 = 2 \cdot \sum_{i=0}^{\infty} \frac{1}{2^i} = \dots$
- b. $\lambda x \lambda y. 2 \cdot (x + y) = \lambda x \lambda y. 2x + 2y$
- c. $\lambda x.(x, x) = \lambda x.\text{let } y = x \text{ in } (y, y)$

crucial to incorporate the context of computations. In extensional settings, both the context and the method of a computation cannot influence outputs of functions. Thus, the semantic analysis does neither reflect why nor how something is true, only **that** something is true (or false).¹⁴ Therefore, we suggest to match the intensionality of the object language by (at least partially) giving up the assumption of extensionality in the formal language. Thus, we distinguish:

- a. **Functional completeness (weak sense):** Every expression is representable.
- b. **Functional completeness (strong sense):** Every expression is representable and uniquely determined by its extension.

The latter corresponds to the standard interpretation of morphisms as functions in CCCs and typed λ -expressions under extensional $\beta\eta$ -equality. The former corresponds to intensional settings, where multiple morphisms may realize the same input-output behavior (β -equality in λ -calculus). From a linguistic perspective, this distinction is not merely technical. It reflects a fundamental property of natural language: expressions are often of intensional nature and, therefore, precisely not uniquely determined by their extension. Interestingly, abandoning (2) for intensional settings then does not constitute a disadvantage or loss of expressivity, but a closer match between object language and formal language. Language rarely represents things in the world uniquely. More often than not, there are multiple (if not infinite) equivalent ways to describe how things in the world are or could be. Thus, representability without uniqueness, i.e., functional completeness in the weak sense, matches the potential ambiguities of co-extensional terms in intensional settings. Morphisms adhering to the weak sense of functional completeness correspond to representations in the conceptual sense, but not functions. In contrast, representability with uniqueness, i.e., functional completeness in the strong sense, then models extensional settings, where co-extensional terms are expected to be equal. Only these morphisms correspond to canonical representations and thus to functions in the classical sense.

Distinguishing between a weak and a strong sense of functional completeness then necessarily imposes a more nuanced notion of identity: A strong version based on co-extensionality alone and a weaker version compatible with intensional settings. This is especially significant, as identity in the weak sense should be more comprehensive than co-extensionality alone, but also not completely arbitrary, and thus vacuous. This motivates a refinement of the notion of identity in intensional domains and category theory provides just that, cf. [24].

¹⁴ The idea that a semantic analysis should represent how things are true (or false) constitutes the core motivation of truthmaker semantics, cf. [11,12], and [3]. As it does not straightforwardly fall into one of the domains of CHLC, a separate discussion of the notions of truthmakers and functional extensionality would be needed.

3 Category Theory for Natural Language Semantics

3.1 A Semantic Category

Since its origins in the 1940s, the main achievement of category theory, sometimes called the science of (mathematical) analogy, probably consists in providing a level of abstraction allowing to generalize notions over different domains, previously only applicable in one domain. Category theory replaces sets and functions on them with the general notion of objects and morphisms between them. As objects remain 'anonymous', there is no usual point-wise equality relation available. While this may initially seem like a disadvantage, it was exact equality that constituted the problem core in intensional domains. We recall that a category $\mathcal{C}$ consists of a collection of objects together with morphisms between them, closed under identity and associative composition, cf. [18,26].

In the linguistic scenario, objects are types (or products of these, i.e., tuples) and morphisms are the meaning terms between these. The compositionality of morphisms corresponds to the compositionality of meaning terms, for example, *Imane Khelif is female* is represented by a composite morphism p structurally depicting the proposition with $p : \mathbf{0} \rightarrow \mathbf{t}, p = f \circ k, k : \mathbf{0} \rightarrow \mathbf{e}$ as the morphism corresponding to the denotation of *Imane Khelif*, and *(is) female* represented by $f : \mathbf{e} \rightarrow \mathbf{t}$.¹⁵ However, as the discussion in the previous sections has shown, linguistic meaning is not exhausted by the restriction of semantic analysis to a strong sense of functional completeness. The same extensional behavior may correspond to distinct conceptual or cognitive structures. This suggests that the failure of extensionality should not be understood as a defect of the logical system but rather as a consistent and welcome property of the semantics in intensional domains. To make this precise, we propose to distinguish between two ontological domains within our semantic category: an extensional domain $\mathbf{0}$ with entities of type $\mathbf{e}$ and predicates on them fully adhering to functional extensionality and an intensional domain $\mathbf{i}$ with entities of type $\mathbf{e}^*$ and predicates on these.

This distinction is also reflected in the introduction of semantic objects into their domains. We assume a canonical way of introducing extensional entities via morphisms $\mathbf{0} \rightarrow \mathbf{e}$, where $\mathbf{0}$ represents an extensional source of reference, formally an initial object with respect to the whole category. Such morphisms correspond to expressions that presuppose the existence of their referents and serve as arguments of extensional predicates. For example, the utterance *Imane Khelif is female* can only be evaluated if there is a morphism $khelif : \mathbf{0} \rightarrow \mathbf{e}$ that corresponds to the referent *Imane Khelif*. In this sense, the introduction of extensional entities is also inherently tied to presuppositional commitments.

In contrast, $\mathbf{i}$, an initial object for the intensional subcategory, represents an

¹⁵ Interestingly, this approach to propositions embodies a middle ground between absolute notions, such as one based on possible worlds, cf. [17], and structured ones, cf. [8]. The composing morphisms are still visible in the diagram, but the composite morphism ties them into a whole with the uniform mechanism of composition, thereby avoiding the problem of the unity of the propositions, cf. [21]. However, it is necessary to differentiate between the diagram and the morphisms in it, cf. [27].

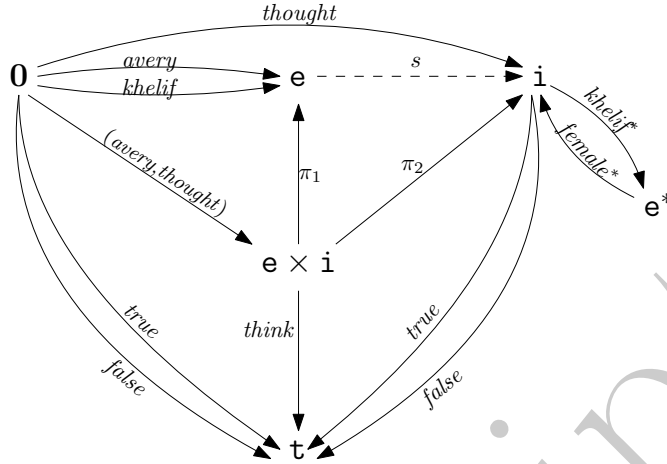


Fig. 1. Part of a semantic category with meaning terms for (8-a).

intensional ontology, not only encompassing the intensional entities e^* , but generally all the kinds of objects that characterize belief, thought, desire, imagination, etc. Intensional entities are introduced via morphisms $i \rightarrow e^*$. These also include hypothetical or non-existent entities and are to be understood as potentially distinct representations of the same extension. In examples like (8-a), meaning terms embedded under intensional predicates such as *think* are interpreted non-extensionally, that is, with a weak notion of functional completeness. The thought object *Imane Khelif* is available as an element of type e^* , but it does not carry an existential commitment and, moreover, *female* is not predicted to apply in an actual sense (since the predicate *think* is not factive).¹⁶ This difference in introduction mechanisms naturally captures both the distinction between extensional and intensional contexts and presuppositional behavior.

Within this setting, predicates can be analyzed as morphisms with arguments in the form of types of both domains. For example, in an extensional setting the predicate *female* is understood as a morphism $female : e \rightarrow t$ and as $female^* : e^* \rightarrow i$ in an intensional setting.¹⁷ Extensional transitive verbs are modeled as morphisms of type $g : (X, e) \rightarrow t$, where X is the type of the logical subject (usually e). Intensional transitive verbs, on the other hand, are modeled as morphisms $g^* : (X, i) \rightarrow t$.¹⁸

¹⁶ As content embedded under factives such as *know* or *regret* must be true, additional constraints are needed to capture the behavior of this particular class of intensional verbs. Cf. 3.3 for a general discussion of constraints, and [27] for factives in particular.

¹⁷ Note that we intentionally do not use t as the codomain for the intensional predicate g^* , as t is reserved for actual truth or falsity. Predicates in intensional settings are not assumed to be true or false in the same sense. This is especially important as intensional domains may embed hypothetical or even contradictory content.

¹⁸ We use i instead of e^* for intensional objects with extra structure, e.g. propositions.

The crucial point is that these two classes of morphisms are subject to different structural constraints. For extensional predicates, we assume that the strong form of functional completeness holds. That is, every expression incorporating objects of type $\mathbf{e}$ can be represented by a morphism, and this representation is unique up to extensional equality. In particular, these morphisms are fully determined by their behavior on arguments: if two morphisms agree on all inputs, they are identical. This reflects the fact that, in the extensional domain, objects are individuated by their extension, and no further structure is relevant for semantic composition. By contrast, intensional predicates are not subject to this constraint. While it is still possible to represent operations on $\mathbf{e}^*$, such representations are no longer unique. Two distinct morphisms $g_1^*, g_2^* : (X, \mathbf{e}^*) \rightarrow \mathbf{t}$ may be identified with the same extensional truth conditions for all arguments, while differing conceptually or in their internal structure. This is not a defect, but a direct consequence of the assumed non-extensional structure of the intensional domain. Since entities of type $\mathbf{e}^*$ encode conceptual or epistemic distinctions that go beyond their extension, it is expected that different morphisms may exploit this structure in different ways, even if their observable behavior coincides extensionally.

The relationship between the two domains of $\mathbf{0}$ and $\mathbf{i}$ is mediated by exactly these objects that induce the intensionality, namely *belief*, *desire*, *thought*, *fear*, etc. These objects, referred to as attitudinal or modal objects by Moltmann [21,22], provide the ground to introduce entities and events in the intensional domain of $\mathbf{i}$. Their content is characterized by endomorphisms $c^* : \mathbf{i} \rightarrow \mathbf{i}$ (e.g., for $khelif^* : \mathbf{i} \rightarrow \mathbf{e}^*$, $female^* : \mathbf{e}^* \rightarrow \mathbf{i}$, we get $p^* = female^* \circ khelif^*$ for the intensional content *Imane Khelif is female* under, e.g., a thought), and as they are truth-evaluable or satisfiable, cf. [22], we distinguish the corresponding morphisms *true*, resp. *false*: $\mathbf{i} \rightarrow \mathbf{t}$, representing meaning terms for expressions like *true belief* or *false hope*, where the truth predicates apply to the content of *belief*, *hope*, etc., but not to the matrix predicate.

Additionally, the intensional abstract object, represented by a morphism $\mathbf{0} \rightarrow \mathbf{i}$, provides the ground for an implicit 'shift' of extensional entities $\mathbf{e}$ to intensional entities $\mathbf{e}^*$ via a dependent factoring morphism $s : \mathbf{e} \rightarrow \mathbf{i}$. It should be noted that this is not a general shift such as the introduction of $\langle s, \alpha \rangle$ for any extensional type α in λ -calculus, as that shift in the category is only implicit and is mediated by and dependent on the existence of a corresponding intensional abstract object such as *belief*. The morphism s is taken to be uninterpretable in the absence of an abstract object with domain $\mathbf{0}$ and codomain $\mathbf{i}$. In its presence, it can embed extensional objects into the intensional domain.¹⁹ Intuitively, this corresponds to viewing an object not merely as a referent but as a conceptual object. This shift has two important consequences. First, it removes presuppositional commitments: an object that was previously taken to be an instance of $\mathbf{e}$ may be considered in $\mathbf{e}^*$ without requiring that its existence be asserted or

¹⁹ The entities of type $\mathbf{e}^*$ do not need to originate from the ones of type $\mathbf{e}$, as $\mathbf{i}$ allows impossible or hypothetical objects, such as *the triangular circle* or *this perpetuum mobile*, which are not expected to exist in an extensional sense.

verified in the actual world. Second, it introduces the possibility of multiple distinct representations of what is, extensionally, the same object. Once embedded in $\mathbf{e}^*$, an object may display different conceptual structures, corresponding to different contextual influences. Importantly, this shift is not reversible. There is no canonical morphism from $\mathbf{i}$ or $\mathbf{e}^*$ back to $\mathbf{e}$, since not every intensional object corresponds to an extensional referent. This asymmetry reflects the familiar observation that while every real object can be thought of, not every object of thought exists. In category theoretic terms, the absence of a morphism $\mathbf{e}^* \rightarrow \mathbf{e}$ also prevents us from isomorphically collapsing the intensional domain into the extensional one, and thereby preserves the distinction between the two.

From the perspective of functional completeness, this two-domain architecture allows us to localize the role of extensionality. In the extensional domain $\mathbf{0}$ with entities of type $\mathbf{e}$ and predicates on them, functional completeness holds in the strong sense: every operation is representable and uniquely determined. In the intensional domain $\mathbf{i}$ with entities $\mathbf{e}^*$ and predicates over them, only the weaker form of functional completeness applies: operations may still be representable, but their representation is not unique. Therefore, an extensional subcategory adheres to functional completeness in the strong sense, and another subcategory, which signifies an intensional domain, only adheres to functional completeness in the weak sense. Crucially, this does not affect the expressive power of the system. The same range of logical operations can be defined in both domains. What differs is the level at which distinctions between morphisms are made.

The resulting picture is one in which the underlying logical system remains unchanged, while the ontological domain determines whether extensionality, and hence unique representability, applies. Extensionality is not a global property of the logic, but a local property of certain types of objects. This allows us to reconcile the somewhat abstract nature of category theory with the empirical reality of natural language, where meaning is often intensional and cannot be reduced to extension alone.

In the previous discussion of (hyper-)intensional phenomena, we have seen how contextual parameters such as perspective or subjective criteria of identity turned out to be crucial in determining meaning up to a conceptual level of meaning. However, until now, we have not said anything about how computation contexts are included in meaning composition. In what follows, we will discuss the significance of the Yoneda lemma for a semantic category.

3.2 Meaning and the Yoneda Perspective

The foregoing discussion has established that intensional objects in the domain of $\mathbf{i}$ resist unique representation: multiple distinct morphisms may share the same extensional behavior while differing in conceptual or epistemic structure. In view of the failure of extensional identity notions in the intensional domain, category theory provides a principled structural concept of identity through the Yoneda lemma. In set-theoretic semantics, equality of objects is defined element-wise: two sets M, M' are equal iff $\forall x : x \in M \Leftrightarrow x \in M'$. Category theory does not provide this notion of equality in general, as objects remain 'anonymous'. They

have no intrinsic properties accessible independently of the morphisms in which they participate. The relevant notion of identity is therefore not strict equality but isomorphy, cf. [20]. Crucially, isomorphisms between objects need not be unique, transforming the question of identity from a binary into a qualitative one: identity then is identity relative to a context of relevant structural relations, matching the intuition that, in the real world, things are virtually never exactly equal but equal enough w.r.t. relevant properties. This relational perspective on identity is made formally precise by the Yoneda lemma.²⁰ The Yoneda lemma formally encodes what we may informally call the **Yoneda perspective**:

(9) **Yoneda perspective**

An object is fully determined by the totality of its relations to all other objects in the category.

Put simply, only the objects in structurally indistinguishable relations to every other object in the category are, from the category's internal perspective, identical. It is the context that determines the relevant qualities of identity.

For the semantic category introduced in section 3.1, this perspective has immediate consequences. In the extensional domain $\mathbf{0}$, no additional context is required, as the strong sense of functional completeness already guarantees unique representability. In this domain, the Yoneda lemma is, so to speak, informationally redundant: it confirms what functional completeness already established. The philosophically and linguistically interesting case is the intensional domain $\mathbf{i}$. Here, the representable functor $\mathcal{C}(\mathbf{i}, -)$ associated to an intensional object encodes not just its truth-conditional behavior, but the full profile of morphisms in which it participates, including morphisms that distinguish structurally different representations of the same extension. Consider again example (7), where the object *Theseus' starship*, once shifted to type $\mathbf{e}^*$ via the factoring morphism s , participates in different morphisms depending on whose perspective structures the intensional domain. Watto's criteria of identity and Obi-Wan's criteria may differ, corresponding to non-isomorphic representable functors over the relevant subcategory. The Yoneda lemma then guarantees that these expressions are genuinely distinct within the intensional subcategory, without requiring us to explicitly enumerate the distinguishing features as additional input parameters of extensional functions.

This is the decisive formal advantage over the extensional strategy described in section 2.2. The classical approach to hyperintensionality is forced to proliferate input parameters in order to break unwanted co-extensional identifications. The Yoneda perspective inverts this: rather than enriching the input space of functions, it uses the relational context in which objects of meaning are individuated. Meaning is not determined by outputs of extensional functions, but by how an object relates to every other object in its semantic context. This constitutes a shift from a functional to a genuinely structural conception of semantic identity. Furthermore, the Yoneda perspective motivates an epistemic rather than realistic notion of truth. In classical formal semantics, truth adheres to an absolute con-

²⁰ Cf. Theorem 2.2.4 in [26].

cept, given solely as a question of membership in sets of possible worlds. Under the Yoneda perspective, the relevant question shifts towards which relational contexts a meaning computation is situated in. This naturally corresponds to the way speakers actually evaluate intensional reports: not by checking absolute extensions, but by situating an expression within a network of inferential and epistemic relations. For (8), for instance, whether *Imane Khelif is female* is true under Avery's thought is not determined by a fixed extension of *female*, but by the relational structure that Avery's intensional domain makes available for the morphism $female^* : \mathbf{e}^* \rightarrow \mathbf{i}$, e.g. the use of a sociological stance. Different relational contexts yield, via the Yoneda lemma, genuinely distinct representations in the intensional domain, even where co-extensionality would identify them. However, it is worth noting that the Yoneda perspective does not make identity arbitrary or context-dependent in unconstrained ways. The natural isomorphism condition is mathematically precise: objects are Yoneda-identical iff they stand in exactly the same morphism-profile to every object in the category. The context-sensitivity arises not from loosening this criterion, but from the fact that the available morphisms within the intensional subcategory induced by attitudinal or modal objects are allowed to vary conceptually. Identity is structural, uniform, and compositional; it is simply relativized to the appropriate domain.

3.3 Enriching Constraints

The semantic category developed in sections 3.1 and 3.2 provides a general ontological architecture and a structural notion of identity grounded in the Yoneda perspective. What remains to be specified is the mechanism by which concrete semantic distinctions are formally encoded within this architecture. In this section, we sketch how category theory allows us to impose such distinctions through morphism-level constraints, and we discuss how these constraints can themselves be given a category theoretic characterization.

The central advantage of a category theoretic setting over a purely set-theoretic one is that it becomes possible to characterize the behavior of semantic operations not by inspecting the internal constitution of objects, but by specifying conditions on the morphisms between them. Such conditions, or constraints, restrict which morphisms are admissible within a given part of the category, thereby encoding semantic distinctions at the level of structure rather than extension. We define two foundational constraints to distinguish the two senses of functional completeness established in section 2.3.

- (10) **C1: Strong functional completeness**
 $\forall f, g \in Hom(\tau, \mathbf{t}) \quad \forall x : f \circ_{\text{ext}} x = g \circ_{\text{ext}} x \Rightarrow f =_{\text{ext}} g$ with $\mathbf{i}$ not in τ
- (11) **C2: Weak functional completeness**
 $\forall f, g \in Hom(\tau, \mathbf{t}) \quad \forall x : f \circ_{\text{ext}} x = g \circ_{\text{ext}} x \not\Rightarrow f =_{\text{ext}} g$
 with $\mathbf{i}$ in τ at least once

Additionally, a particularly powerful aspect of the category theoretic framework is the possibility to reason about function evaluation at the same level as func-

tions themselves, by considering exponential objects. In CCCs, for any two objects A, B , the exponential object B^A represents the object of morphisms from A to B , together with an evaluation morphism $\text{ev}: B^A \times A \rightarrow B$. Constraints on semantic operations can therefore be expressed as conditions on morphisms through exponential objects, rather than as metalinguistic stipulations external to the category. This is particularly relevant for the distinction between the two classes of predicates. A predicate in the extensional domain corresponds to a morphism $g: \mathbf{e} \rightarrow \mathbf{t}$, while its intensional counterpart corresponds to $g^*: \mathbf{e}^* \rightarrow \mathbf{i}$. The exponential objects $\mathbf{t}^{\mathbf{e}}$ and $\mathbf{i}^{\mathbf{e}^*}$ then represent the spaces of extensional and intensional predicates, respectively. The constraints (C1) and (C2) can thus be understood as a condition on the morphisms into these exponential objects: those factoring through $\mathbf{t}^{\mathbf{e}}$ are subject to strong functional completeness, while those factoring through $\mathbf{i}^{\mathbf{e}^*}$ are not. Importantly, this means the hom-sets as the objects of constraints are internal to the category, not external to its logic. Finally, we note that the constraints themselves might admit a category theoretic description. Rather than treating them as external stipulations on an otherwise unconstrained category, one can consider enriched categories, in which the hom-sets $\mathcal{C}(a, b)$ are not merely sets but objects in some monoidal category $\mathcal{V}$. Replacing hom-sets with structured objects allows additional information, such as the strength of an entailment, the degree of perspective-dependence, or the type of modal operator, to be encoded in the morphism structure itself. Under this view, the constraints (C1)–(C2) would not be imposed from outside the category but would follow from the enriched structure of its hom-objects itself. We leave a full development of this idea for future work, but note that it supports the broader ambition of this paper: to provide a uniform, compositional, and fully internalized category theoretic account of intensional meaning.

4 Conclusion

This paper has examined the role of functional extensionality in CHLC and its consequences for natural language semantics. We have argued that the extensionality assumption, inherited implicitly by both STTs and RTTs, is not a neutral technical choice but a substantive commitment that creates systematic friction with the intensional and hyperintensional dimensions of meaning. In response, we have proposed a two-domain semantic category that distinguishes an extensional domain, in which functional completeness holds in the strong sense, from an intensional domain, in which only the weaker requirement of representability applies. This architecture preserves compositionality while accommodating the fact that co-extensional expressions can differ in meaning. The Yoneda lemma provides the positive notion of identity appropriate to the intensional domain: objects are individuated structurally, through their full profile of relations to other objects in the category, rather than pointwise through their extensions. Finally, we have sketched how semantic constraints can be encoded as morphism-level conditions internal to the category, with enriched categories offering a natural direction for future development. The broader upshot is that

the category theoretic dimension of CHLC, long overshadowed by its logical and computational sides, offers genuine and largely untapped resources for formal semantics. Treating meaning structurally and relationally is not a retreat from formal rigor but a more faithful alignment between the formal language and the (intensional) object language it is meant to describe.

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